

Question	Category	Resource Link
Design URL shortener	Data modeling + Hashing	https://github.com/donnemartin/system-design-primer
Design a pastebin/service for sharing text files	Storage + Expiry + Security	https://github.com/donnemartin/system-design-primer
Design a scalable web crawler	Distributed systems + Scheduling	https://www.geeksforgeeks.org/system-design/system-design-tutorial/
Design Twitter (feed/timeline)	Fan-out, feeds, data model	https://github.com/donnemartin/system-design-primer
Design Instagram (photo sharing)	Media storage, CDNs, thumbnails	https://github.com/donnemartin/system-design-primer
Design YouTube (video streaming)	Media transcoding, CDN, streaming	https://github.com/donnemartin/system-design-primer
Design a chat application (1:1 and group)	Messaging queues, delivery guarantees	https://www.geeksforgeeks.org/system-design/system-design-tutorial/
Design a notification system	Push services, fan-out, retries	https://github.com/donnemartin/system-design-primer
Design a real-time collaborative editor	Operational transform / CRDTs	https://github.com/donnemartin/system-design-primer
Design an online multiplayer game backend	Realtime messaging, state sync	https://www.geeksforgeeks.org/system-design/system-design-tutorial/
Design a ride-hailing service (like Uber)	Geo-partitioning, matching, real-time	https://github.com/donnemartin/system-design-primer
Design a delivery system (logistics)	Routing, scheduling, state management	https://www.geeksforgeeks.org/system-design/system-design-tutorial/
Design a restaurant reservation system	Availability, concurrency control	https://www.geeksforgeeks.org/system-design/system-design-tutorial/
Design an e-commerce website	Catalog, shopping cart, payments	https://github.com/donnemartin/system-design-primer
Design an order processing pipeline	Message queues, idempotency, retries	https://github.com/donnemartin/system-design-primer
Design a search engine (basic)	Indexing, sharding, ranking	https://github.com/donnemartin/system-design-primer
Design autocomplete for search	Trie / prefix indexes, caching	https://www.geeksforgeeks.org/system-design/system-design-tutorial/
Design rate limiter for APIs	Token bucket / leaky bucket, distributed	https://www.geeksforgeeks.org/system-design/system-design-tutorial/
Design a distributed cache	Cache invalidation, replication, consistency	https://github.com/donnemartin/system-design-primer
Design a key-value store	Data partitioning, replication, consistency	https://github.com/donnemartin/system-design-primer
Design a relational database at scale	Sharding, replication, transactions	https://www.geeksforgeeks.org/system-design/system-design-tutorial/
Design a time-series database	Write-heavy patterns, compression, retention	https://github.com/donnemartin/system-design-primer
Design a message queue system (like Kafka)	Partitioning, retention, consumers	https://github.com/donnemartin/system-design-primer
Design a task scheduler (cron at scale)	Distributed scheduling, failure handling	https://www.geeksforgeeks.org/system-design/system-design-tutorial/
Design a metrics and monitoring system	Collection, aggregation, alerting	https://github.com/donnemartin/system-design-primer
Design a logging pipeline	Ingestion, storage, search (ELK-style)	https://github.com/donnemartin/system-design-primer
Design a system for storing user profiles	Schema design, privacy, indexing	https://www.geeksforgeeks.org/system-design/system-design-tutorial/
Design a recommendation system (basic)	Offline training, online serving, features	https://github.com/donnemartin/system-design-primer
Design a notification email system	Deliverability, bounce handling, throttling	https://www.geeksforgeeks.org/system-design/system-design-tutorial/
Design a full-text search service (e.g., for documents)	Indexing, ranking, shard placement	https://github.com/donnemartin/system-design-primer
Design a photo storage service with thumbnails	Object storage, background processing	https://github.com/donnemartin/system-design-primer
Design a content delivery network (CDN)	Edge caching, origin fetch, invalidation	https://aws.amazon.com/architecture/
Design a distributed lock service	Consensus, leader election, fencing	https://github.com/donnemartin/system-design-primer
Design an analytics event pipeline	Event collection, buffering, aggregation	https://github.com/donnemartin/system-design-primer
Design a billing system	Transactions, reconciliation, audit trails	https://www.geeksforgeeks.org/system-design/system-design-tutorial/
Design a system to detect spam or abusive content	ML inference, feature pipelines, throttling	https://github.com/donnemartin/system-design-primer
Design a document collaboration platform (Google Docs-like)	Realtime sync, conflict resolution	https://github.com/donnemartin/system-design-primer
Design a system for feature flags	Configuration, rollout, targeting	https://www.geeksforgeeks.org/system-design/system-design-tutorial/
Design a distributed search for social feed	Realtime indexing, personalization	https://github.com/donnemartin/system-design-primer
Design a schema migration service	Versioning, backfills, zero-downtime deploys	https://www.geeksforgeeks.org/system-design/system-design-tutorial/
Design a file upload service at scale	Chunking, resumable uploads, storage	https://github.com/donnemartin/system-design-primer
Design a push notification system for mobile	Delivery guarantees, platform specifics	https://www.geeksforgeeks.org/system-design/system-design-tutorial/
Design a URL routing layer for microservices	Ingress, API gateway, service discovery	https://aws.amazon.com/architecture/
Design a system to handle webhooks reliably	Delivery guarantees, retry, deduplication	https://github.com/donnemartin/system-design-primer
Design an image recognition pipeline	Batch vs real-time inference, pipelines	https://github.com/donnemartin/system-design-primer
Design a machine learning feature store	Consistency, offline/online features	https://github.com/donnemartin/system-design-primer
Design a system for A/B testing	Experimentation platform, metrics, rollout	https://www.geeksforgeeks.org/system-design/system-design-tutorial/
Design a scalable authentication system (SSO)	Tokens, session stores, OAuth flows	https://www.geeksforgeeks.org/system-design/system-design-tutorial/
Design a single sign-on (SSO) / identity service	Federation, SSO protocols, security	https://www.geeksforgeeks.org/system-design/system-design-tutorial/
Design a content moderation pipeline	Queues, human-in-loop, ML triage	https://github.com/donnemartin/system-design-primer
Design a notification center (multi-channel)	Fan-out, channel adapters, preferences	https://github.com/donnemartin/system-design-primer
Design a system to stream live video	Low-latency streaming, ingestion, transcoding	https://github.com/donnemartin/system-design-primer
Design an IoT telemetry ingestion system	High-throughput ingestion, TTL, compression	https://github.com/donnemartin/system-design-primer
Design a distributed graph storage	Graph partitioning, traversal performance	https://www.geeksforgeeks.org/system-design/system-design-tutorial/
Design a graph-based social network recommendation engine	Graph algorithms, offline/online pipelines	https://github.com/donnemartin/system-design-primer
Design an SMS service gateway	Throughput, carrier integration, retries	https://www.geeksforgeeks.org/system-design/system-design-tutorial/
Design a user activity feed with personalization	Feed ranking, ML features, caching	https://github.com/donnemartin/system-design-primer
Design a scalable API rate-limiting gateway	Distributed counters, token buckets	https://www.geeksforgeeks.org/system-design/system-design-tutorial/
Design a backup and restore system	Snapshots, incremental backups, consistency	https://aws.amazon.com/architecture/
Design a system to perform ETL at scale	Batch/stream processing, orchestration	https://github.com/donnemartin/system-design-primer
Design a fraud detection system	Real-time rules, ML scoring, alerting	https://github.com/donnemartin/system-design-primer
Design a distributed configuration service	Consistency, rollout, versioning	https://www.geeksforgeeks.org/system-design/system-design-tutorial/
Design a system for capacity planning and autoscaling	Metrics-driven scaling, policies	https://aws.amazon.com/architecture/
Design a service mesh for microservices	Sidecars, control plane, observability	https://github.com/donnemartin/system-design-primer
Design an event sourcing system	Event store, replayability, materialized views	https://github.com/donnemartin/system-design-primer
Design a CQRS-based system for high-write workloads	Command/query separation, eventual consistency	https://www.geeksforgeeks.org/system-design/system-design-tutorial/
Design a distributed transaction coordinator	Two-phase commit, idempotency, compensation	https://github.com/donnemartin/system-design-primer
Design a global database with low latency	Geo-replication, conflict resolution, routing	https://github.com/donnemartin/system-design-primer
Design a service for geospatial queries (nearby search)	Geo-indexing, spatial partitioning	https://www.geeksforgeeks.org/system-design/system-design-tutorial/
Design an image processing pipeline with thumbnails and filters	Background workers, retries, storage	https://github.com/donnemartin/system-design-primer
Design a polling system for online surveys	Responses collection, dedup, scaling	https://www.geeksforgeeks.org/system-design/system-design-tutorial/
Design a system to handle large file diffs (e.g., git hosting)	Delta storage, chunking, compression	https://github.com/donnemartin/system-design-primer
Design a scalable comment system	Threading, moderation, notifications	https://www.geeksforgeeks.org/system-design/system-design-tutorial/
Design a schema-less document store	Indexing, query support, scaling	https://github.com/donnemartin/system-design-primer
Design a streaming analytics dashboard	Real-time metrics, windowing, visualization	https://github.com/donnemartin/system-design-primer